UNIT 1 · STANDARD 6.NS.4

Greatest Common Factor

Lesson 1-2

Name: ____________ Date: ____________ Class: ____________

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

Factors of 12: 1, 2, 3, 4, 6, 12 — each divides 12 with no remainder

Factor

A number that divides another number with nothing left over.

Factors of 12: {1, 2, 3, 4, 6, 12}. Factors of 18: {1, 2, 3, 6, 9, 18}. Shared: 1, 2, 3, 6. GCF = 6

Greatest Common Factor

The biggest number that divides two or more numbers evenly.

8 and 12 both divide evenly by 1, 2, and 4 — those are their common factors

Common factor

A number that divides two or more numbers evenly.

$15 \div 3 = 5$ with no remainder, so 15 is divisible by 3

Divisible

A number you can divide evenly, with nothing left over.

$12 = 2 \times 2 \times 3$ and $18 = 2 \times 3 \times 3$. Shared primes: $2 \times 3 = 6 = GCF$

Prime factorization

Writing a number as prime numbers multiplied together. It helps you find the GCF.

Key Ideas & Notes

🎯 WHAT WE'RE LEARNING TODAY

I can find the greatest common factor (GCF) of two numbers by listing or comparing their factors.

1 Notice

Mission Control needs to organize 24 food packs and 36 water containers into equal groups for the crew pods.

2 Key idea

I can find the greatest common factor (GCF) of two numbers by listing or comparing their factors.

3 Words I'll use

- **Factor**
- **Greatest Common Factor**
- **Common factor**
- **Divisible**
- **Prime factorization**

🤔 Think About It

- What two quantities need to be divided equally?
- What numbers could divide into both 24 and 36?
- What's the biggest number of equal groups we can make?

👀 Watch

Watch your teacher model one example. Jot what you see.

____

🤝 We try

Solve the next one together as a class.

____

____

✏️ You try

Now try one on your own.

____

____

Turn & Talk — Discussion Points

Level 1 support · Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

Mission Control must split 24 food packs and 36 water containers into equal crew pods, with each pod getting the same of each. Why won't the answer just be 24 or 36?

Level 1 support

Start here: The number of pods must divide BOTH 24 and 36 evenly.

💡 **Need a hint?**

Sentence starters (optional)

The number of pods must be a factor of both ____ and ____.

El número de cápsulas debe ser un factor de ____ y de ____.

It cannot be 36 because ____.

No puede ser 36 porque ____.

Word bank: factor · common factor · divisible · greatest common factor · equal groups

Listen for: *Students see the number of pods has to be a common factor of 24 and 36, so it must divide both with nothing left over, ruling out numbers that only divide one of them.*

Level 2

If Mission Control added one more food pack (25 packs and 36 containers), how would that change the greatest number of equal pods? Justify your reasoning.

- With 25 and 36 the GCF becomes ____ because ____.
- Changing one number changes the shared factors, so ____.

EXPLORE

How did you find the factors that 24 and 36 share, and how did you decide which one is the GCF?

Level 1 support

Start here: The GCF is the largest factor that appears in both lists.

💡 **Need a hint?**

Sentence starters (optional)

The common factors are ___.

Los factores comunes son ___.

The greatest common factor is ___ because ___.

El máximo común divisor es ___ porque ___.

Word bank: factor | common factor | greatest common factor | divisible | prime factorization

Listen for: *Students list common factors 1, 2, 3, 4, 6, 12 of 24 and 36 and identify 12 as the GCF because it is the largest factor shared by both.*

Level 2

Could you find the GCF of 24 and 36 with prime factorization instead of listing factors? Show how the shared primes give 12.

- Using primes, 24 = ___ and 36 = ___, so the shared primes are ___.
- Multiplying the shared primes gives ___, which matches the GCF because ___.

CONNECT

The baker has 48 cookies and 32 brownies and wants identical boxes with none left over. How does the GCF tell her the most boxes she can make?

Level 1 support

Start here: The GCF of 48 and 32 is the greatest number of equal boxes.

💡 **Need a hint?**

Sentence starters (optional)

The baker can make ___ boxes because the GCF of 48 and 32 is ___.

La panadera puede hacer ___ cajas porque el MCD de 48 y 32 es ___.

Each box gets ___ cookies and ___ brownies.

Cada caja recibe ___ galletas y ___ brownies.

Word bank: **greatest common factor** **factor** **divide** **equal** **remainder**

Listen for: *Students compute GCF(48, 32) = 16, so 16 boxes, each with 3 cookies and 2 brownies, and explain why a bigger number would leave items over.*

Level 2

Why does the baker use the GREATEST common factor instead of just any common factor like 8? What changes if she picks a smaller one?

- She uses the greatest common factor because ___.
- If she used 8 instead, she would make ___ boxes, but ___.

REFLECT

How is the greatest common factor different from the least common multiple, and how can you tell which one a problem needs?

Level 1 support

Start here: GCF splits things into equal groups; it is the largest shared factor.

💡 **Need a hint?**

Sentence starters (optional)

The GCF is the largest number that ____, but the LCM is the smallest number that ____.

El MCD es el número más grande que ____, pero el mcm es el número más pequeño que ____.

A problem needs the GCF when ____.

Un problema necesita el MCD cuando ____.

Word bank: greatest common factor | common factor | divisible | factor | equal groups

Listen for: *Students contrast GCF (largest factor that divides both, used to split into equal groups) with LCM (smallest shared multiple) and connect GCF to making equal groups with nothing left over.*

Level 2

Two numbers are relatively prime (GCF = 1). What does that tell you about splitting them into equal groups? Give an example.

- If the GCF is 1, then the only equal grouping is ____ because ____.
- An example is ____ and ____, which can only be split into ____.

Guided Examples

Example 1

What is the GCF of 8 and 12?

Solution: Factors of 8: 1, 2, 4, 8. Factors of 12: 1, 2, 3, 4, 6, 12. The largest common factor is 4.

Answer: A. 4

Example 2

Which pair of numbers has a GCF of 5?

Solution: $15 = 3 \times 5$, $25 = 5 \times 5$. Both share the factor 5, and 5 is the largest shared factor.

Answer: A. 15 and 25

Example 3

What is the GCF of 14 and 21?

Solution: Factors of 14: 1, 2, 7, 14. Factors of 21: 1, 3, 7, 21. The largest common factor is 7.

Answer: A. 7

Write About the Math

The Writing Revolution

I can explain how I found the GCF using the words factor, common factor, divisible, and greatest common factor.

1. Kernel Sentence

subject + verb

Model: Factor is a number that divides another number with nothing left over.
Factor es un número que divide a otro sin que sobre nada.

Write a kernel sentence about factor. Use a subject and a verb.
Escribe una oración base sobre factor. Usa un sujeto y un verbo.

____

____

2. Sentence Expansion

because · but · so

Kernel: Factor matters in math
Factor importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because *porque*

Factor matters in math because ____.
Factor importa en matemáticas porque ____.

____

but *pero*

Factor matters in math, but ____.
Factor importa en matemáticas, pero ____.

____

so *entonces*

Factor matters in math, so ____.
Factor importa en matemáticas, entonces ____.

____

3. Sentence Types

4 ways to write a math idea

Statement
Afirmación

Tell one true fact about factor.

Di un hecho verdadero sobre factor.

Factor ____.

Question
Pregunta

Ask a question about factor.

Haz una pregunta sobre factor.

How does ____ ?

¿Cómo ____ ?

Exclamation
Exclamación

Show excitement about factor.

Muestra entusiasmo sobre factor.

Wow, ____ !

¡Guau, ____ !

Command
Mandato

Tell a partner what to do with factor.

Dile a un compañero qué hacer con factor.

First, ____ .

Primero, ____ .

4. Explain Your Reasoning

use a sentence starter

The number of pods must be a factor of both ____ and ____.

El número de cápsulas debe ser un factor de ____ y de ____.

It cannot be 36 because ____.

No puede ser 36 porque ____.

The common factors are ____.

Los factores comunes son ____.

Try It

Solve on your own. Check the answer key when you are done.

1. What is the GCF of 9 and 15?

A. 3

B. 1

C. 9

D. 45

Show your work:

2. A teacher has 36 pencils and 48 erasers. She wants to make identical supply bags with no items left over. What is the greatest number of bags she can make?

A. 12 bags

B. 6 bags

C. 4 bags

D. 36 bags

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A farmer has 48 apples and 32 oranges. She wants to make gift baskets where every basket has the same number of apples and the same number of oranges, with none left over. Find the greatest number of baskets and explain how you used GCF to solve this.

Sentence starter: The GCF of 48 and 32 is ___ because ___. So the farmer can make ___ baskets, each with ___ apples and ___ oranges.

Show your work:

Reflect — Exit Ticket

What is the GCF of 18 and 27?

A. 9

B. 3

C. 6

D. 18

Your answer:

Answer Key & Teacher Guide

1. **Try It 1:** A. 3 *— Factors of 9: 1, 3, 9. Factors of 15: 1, 3, 5, 15. The largest common factor is 3.*
2. **Try It 2:** A. 12 bags *— GCF(36, 48) = 12. She can make 12 bags, each with 3 pencils and 4 erasers.*
3. **Exit Ticket:** A. 9 *— Factors of 18: 1, 2, 3, 6, 9, 18. Factors of 27: 1, 3, 9, 27. Common: 1, 3, 9. GCF = 9.*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Factor is a number that divides another number with nothing left over.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).